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Electrical properties and crystal structure of solid electrolyte based on lithium hafnium phosphate $LiHf_2(PO_4)_3$

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The electrical properties and the crystal structure were investigated for the ceramic electrolytes based on $LiHf_2(PO_4)_3$. The conductivity enhanced by the Li_2O addition with $LiHf_2(PO_4)_3$ system and by the increase of x for $Li_{1+x}M_xHf_{2-x}(PO_4)_3$ (M=Cr, Fe, Sc, In, Lu or Y) systems. The $P2_1/n$ monoclinic phase transformed to NASICON-type $R\bar{3}c$ rhombohedral phase at above 1173 K. The activation energy for Li^+ ion migration was decreased by the phase transition. The activation energy for bulk component was 0.42 eV for the NASICON-type structure. A maximum conductivity at 298 K is 1.7×10^{-4} S·cm^{-1} for the sample of $Li_{1.2}Fe_{0.2}Hf_{1.8}(PO_4)_3$.

1. Introduction

A lithium battery has an advantage of obtaining a high energy density because of its high electrochemical potential and light weight. A high Li^+ conducting solid electrolyte at room temperature has been investigated as the application for an all solid state lithium battery.

Goodenough et al. [1,2] discovered a Na^+ super ionic conductor of $Na_{1+x}Zr_2P_{3-x}Si_xO_{12}$ (NASICON). $LiM_2(PO_4)_3$, M=Ge [3,4], Ti [5–15], Sn, Hf [15,16], Zr [16,17] also possess the NASICON-type structure. We have already reported that $LiTi_2(PO_4)_3$ has the most suitable lattice size (cell volume ca. 1309 Å^3, space group $R\bar{3}c$) for Li^+ migration in the NASICON-type Li^+ conductors [15]. A minimum activation energy of 0.28–0.30 eV for a bulk component was obtained for the Li^+ conductors based on $LiTi_2(PO_4)_3$. For $LiGe_2(PO_4)_3$ series (cell volume ca. 1218 Å^3) the tunnel size is small for a Li^+ ionic migration [4]. $LiSn_2(PO_4)_3$ and $LiZr_2(PO_4)_3$ series show the lower conductivity, because the rhombohedral structure was distorted to monoclinic one at room temperature. We have reported that the conductivity at around room temperature is mainly controlled by that of grain boundaries for these NASICON-type ceramic electrolytes [15]. The conductivity and the sinterability at the grain boundaries are greatly increased by the excessive lithium content by the addition of lithium compound or by the M^{3+} substitution [4,13–15].

In this paper, the electrical properties and the crystal structure were examined for the NASICON-type solid electrolytes based on $LiHf_2(PO_4)_3$, i.e. $LiHf_2(PO_4)_3+yLi_2O$ system and $Li_{1+x}M_xHf_{2-x}(PO_4)_3$ (M=Cr, Fe, Sc, In, Lu or Y) system.

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2. Experimental

2.1. Materials

Li_2CO_3 (purity: 99.99%), HfO_2 (99.9%), $(NH_4)_2HPO_4$ (extra pure grade), and M_2O_3 M=Cr, Fe, Sc, In, Lu, or Y (99.9% or 99.99%) were used as starting materials. The mixture was reacted in a platinum crucible at 1173 K for two hours in air. The starting materials of Li_2CO_3 and $(NH_4)_2HPO_4$ decomposed into Li_2O and P_2O_5 by the heat treatment. The resulting material was ground into a fine powder using a ball-mill for six hours (wet process: methanol). The dried powder was again reacted at 1173 K for two hours in air and then reground by the same ball-milling method. The particle size of the powder was smaller than 1 μm. Dried powder mixed with a suitable amount of 3 wt% PVA solution was pressed into a pellet at a pressure of 1×10^8 Pa. The pellet was sintered for two hours in air. Gold electrodes were sputtered on both polished surfaces of the electrolyte by an Ion Coater (Shimadzu IC-50).

2.2. Measurement

An X-ray powder diffraction analysis was conducted using a Rigaku Geigerflex with Si powder (99.99%) as an internal standard. Scanning electron micrographic measurements were performed with Jeol JXA-733 X-ray microanalyzer. The conductivity was determined by a complex impedance method using an LCZ meter (10^2–10^6 Hz, 4276A and 4277A, Hewlett Packard Co., Ltd). The apparent porosity was calculated from the apparent density of sintered pellets measured by the Archimedes method.

3. Results and discussion

3.1. Crystal structure of $LiHf_2(PO_4)_3$

A NASICON-type structure for $A(I)M(IV)_2(PO_4)_3$ A=Li, Na, Ag, etc., M=Ge, Ti, Sn, Hf, Zr is composed of MO_6-octahedra and PO_4-tetrahedra. There are four possible sites (one A_1 and three A_2) per formula unit for monovalent cation. The A_1 sites are fully occupied by monovalent cations, and the A_2 sites are vacant. However, the sintering process should be conducted at high temperature to obtain the rhombohedral structure in the case of the solid electrolyte based on $LiHf_2(PO_4)_3$. Fig. 1 presents the X-ray diffraction patterns for the sample of $LiHf_2(PO_4)_3$ sintered at 1393 K and 1493 K. The X-ray pattern for the sample sintered at 1493 K is the same as that for the NASICON R$\bar{3}$c structure. However, the sample sintered at 1393 K is identical with the β-$Fe_2(SO_4)_3$-type monoclinic phase ($P2_1/n$) [18,19].

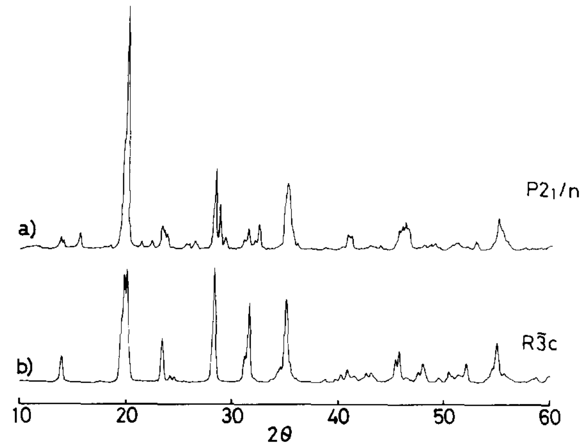

Fig. 1. The X-ray diffraction patterns (Cu Kα) for $LiHf_2(PO_4)_3$ sintered at 1393 K (a) and 1493 K (b).

3.2. $LiHf_2(PO_4)_3+yLi_2O$ system

The relationship between the sintering temperature and the porosity of the sintered pellet is shown in fig. 2. The high density pellet could not be obtained for $LiHf_2(PO_4)_3$ even if the sintering temperature raised to 1593 K. On the other hand, high density pellet can be obtained for the samples with Li_2O addition. Li_2O acts as a flux to form a high density sample. The effect of the Li_2O addition has been already reported in our previous papers [15]. The porosity greatly decreased by the sintering at around 1173 K, and then increased by the elevation of the sintered temperature for all the Li_2O added samples.

Fig. 3 shows the surface microstructure of the sintered $LiHf_2(PO_4)_3+0.3Li_2O$ samples. The grains are closely contacted with one another for the sample heated at 1173 K (fig. 3a). The particle size was smaller than 1 μm, which is almost the same as that

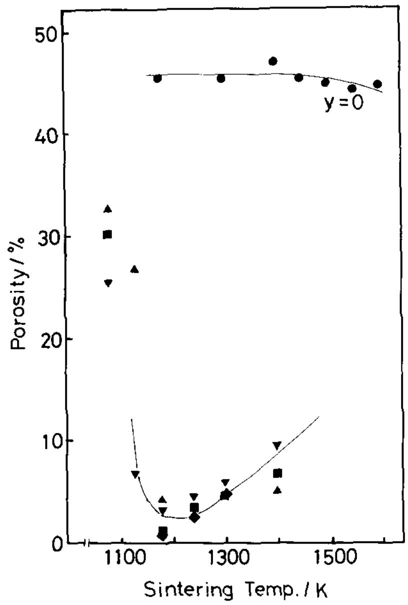

Fig. 2. Relationship between the porosity and the sintered temperature for $LiHf_2(PO_4)_3+yLi_2O$ system: $y=0$ (●); 0.1 (▲); 0.2 (▼); 0.3 (■); 0.4 (♦).

of the ball-milled powder. The surface of the sintered sample is very smooth. No other peaks without those of $LiHf_2(PO_4)_3$ can be detected from the X-ray diffraction patterns for all $LiHf_2(PO_4)_3+yLi_2O$ system. In addition, the lithium atom did not evaporate by the heating process, which was confirmed by an atomic absorption measurement. The added Li_2O would exist as a glassy phase at grain boundaries. However, Li_2O itself has a high melting point. We presume that the glassy lithium phosphate would be formed at elevated temperature from the added Li_2O and phosphate. The grain size grew with the elevation of the sintering temperature (b). The fine granules appeared on $LiHf_2(PO_4)_3$ grains. The granules would be the lithium phosphate or Li_2O which acts as a flux. Some pores observed between the particles appeared in fig. 3b. The increase in the pellet porosity with the sintering temperature (in fig. 2) might result from the fact that the pore size at the grain boundary increased with the particle growth. The particle growth greatly accelerated for the sample sintered at 1393 K (fig. 3c). In this case, cracks are observed in the grains.

Fig. 4 presents the temperature dependence of the total (bulk plus grain boundary component) conductivity for $LiHf_2(PO_4)_3+0.3Li_2O$ sample. The conductivity greatly enhanced for the samples sintered at above 1293 K. The β-$Fe_2(SO_4)_3$-type ($P2_1/n$) monoclinic phase was formed for the samples sintered at temperature lower than 1293 K. The slope of the $\sigma T-1/T$ relation decreased by the formation of the NASICON-type structure. It seems that the Li^+ ion does not easily migrate in the monoclinic phase. Fig. 5 shows the relationship between the activation energy for the total conductivity and the sintering temperature. The activation energy was determined in the temperature range from room temperature to 400 K. The activation energy abruptly decreased with the sintering temperature for all the samples. The decrease in the activation energies is attributed to the $P2_1/n \rightarrow R\bar{3}c$ phase transition. Although the samples calcined at 1173 K during the preparation process, $R\bar{3}c$ phase could not be formed for all the calcined powder. The obtained rhombohedral phase did not revert to the monoclinic phase by cooling. The phase transition temperature is about 1423 K for $LiHf_2(PO_4)_3$ $y=0$, and the temperature decreased with the Li_2O addition. The transition temperature for Li_2O added samples is close to that of the high densification (in fig. 2). The decrease of the transition temperature might be influenced by the high sinterability. Fig. 6 shows the total conductivity at 298 K for $LiTi_2(PO_4)_3+yLi_2O$ system. the conductivity considerably increased with the sintering temperature. It is considered that the conductivity was enhanced by both the formation of the NASICON-type phase and the decrease in the porosity. Although the porosity decreased for the monoclinic samples of $y=0.3$ and 0.4 sintered at 1173 K, the conductivity enhancement with high densification is smaller than that with the formation of NASICON-type phase. The conductivity enhancement mainly occurred by the formation of the NASICON-type phase. The conductivity also greatly enhanced by the Li_2O addition. For Li_2O added samples, a maximum conductivity was obtained for the samples sintered at ca. 1298 K. The conductivity decreased for the samples sintered above 1298 K. The particle growth might affect the decrease of the conductivity (see fig. 3). The maximum conductivity is 1.1×10^{-4} $S \cdot cm^{-1}$ for $LiHf_2(PO_4)_3+0.1Li_2O$ sintered at 1298 K.

Complex impedance plot for the rhombohedral

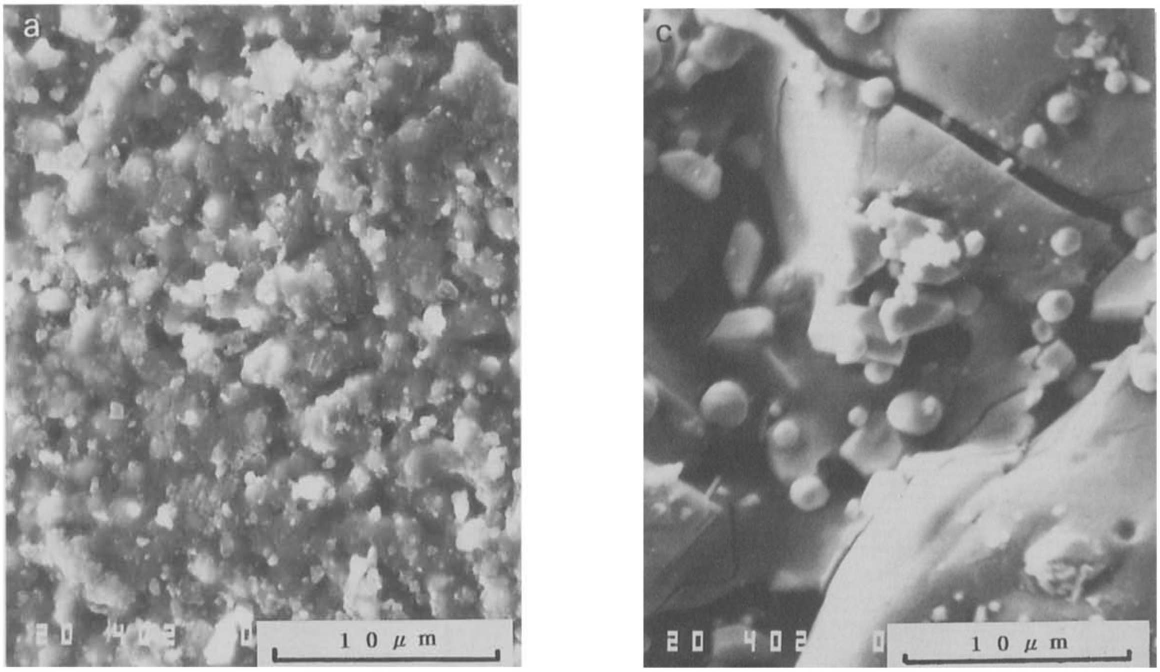

Fig. 3. Surface microstructure for $LiHf_2(PO_4)_3+0.3Li_2O$ sintered at 1173 K (a), 1293 K (b), and 1393 K (c).

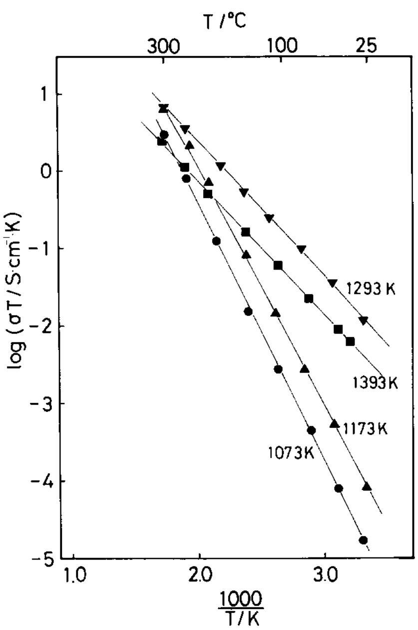

Fig. 4. The $\sigma T-1/T$ relation for $LiHf_2(PO_4)_3+0.3Li_2O$ samples.

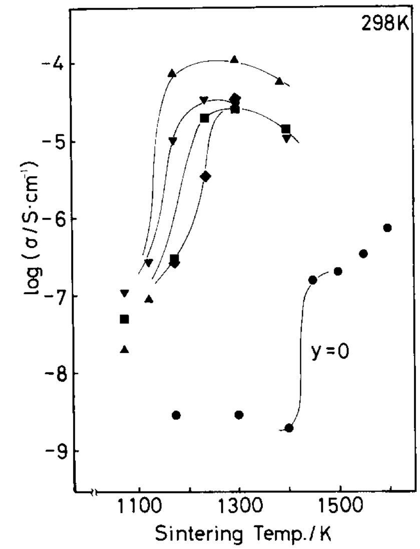

Fig. 6. Relationship between the total conductivity at 298 K and the sintered temperature for $LiHf_2(PO_4)_3+yLi_2O$ system: $y=0$ (●); 0.1 (▲); 0.2 (▼); 0.3 (■); 0.4 (♦).

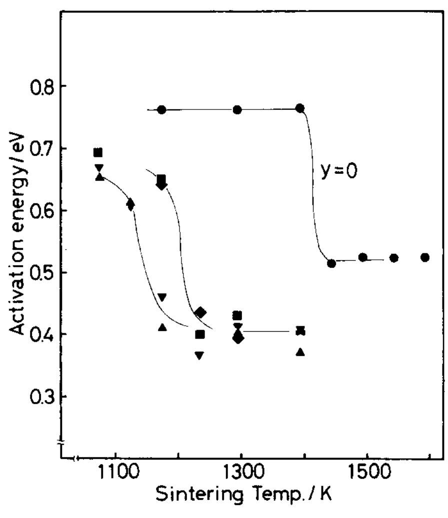

Fig. 5. Relationship between the activation energy and the sintered temperature for $LiHf_2(PO_4)_3+yLi_2O$ system: $y=0$ (●); 0.1 (▲); 0.2 (▼); 0.3 (■); 0.4 (♦).

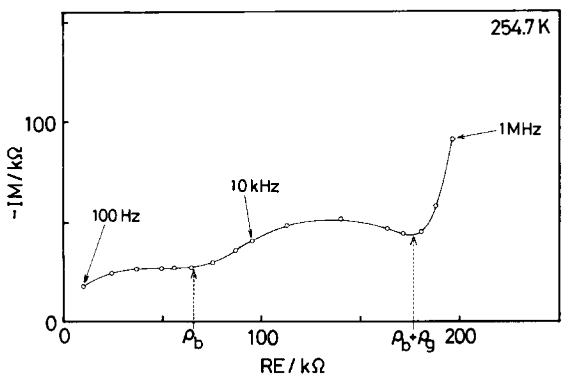

Fig. 7. Cole–Cole plot for $LiHf_2(PO_4)_3+0.1Li_2O$ at 254.7 K.

sample of $LiHf_2(PO_4)_3+0.1Li_2O$ is shown in fig. 7. Two semicircles were obtained for the high frequency region. Two minimum points of the imaginary part in the complex impedance plot are attributed to the bulk and the total (bulk plus grain boundary) resistance, respectively. The conductivities for the bulk and the grain boundary were determined from these results. Table 1 shows the activation energy for Li^+ conduction of the bulk and the grain boundary for $LiHf_2(PO_4)_3+yLi_2O$ (rhombohedral sample). The activation energy for $y=0.1$, 0.2 was determined in the temperature range between 223 K and room temperature. The activation energy at the grain boundary decreased by the Li_2O addition. The enhancement of the conductivity by the Li_2O addition is attributed to the decrease in the ac-

Table 1
The activation energy of the bulk and the grain boundary for $LiHf_2(PO_4)_3+yLi_2O$ system [16].

Sample	Bulk	Grain boundary (eV)
$LiHf_2(PO_4)_3$	0.43	0.56
$LiHf_2(PO_4)_3+0.1Li_2O$	0.42	0.47
$LiHf_2(PO_4)_3+0.2Li_2O$	0.42	0.48

tivation energy at the grain boundary and the high densification. On the other hand, the activation energy for bulk component remains constant (ca. 0.42 eV) for $LiHf_2(PO_4)_3$ network structure. This value is higher than 0.30 eV of $LiTi_2(PO_4)_3$ system (cell volume: 1309 Å^3), because the lattice size for Li^+ migration in $LiHf_2(PO_4)_3$ (cell volume: 1486 Å^3) is larger than that in $LiTi_2(PO_4)_3$. The tunnel size for the 3D structure of $LiHf_2(PO_4)_3$ is too large for Li^+ migration.

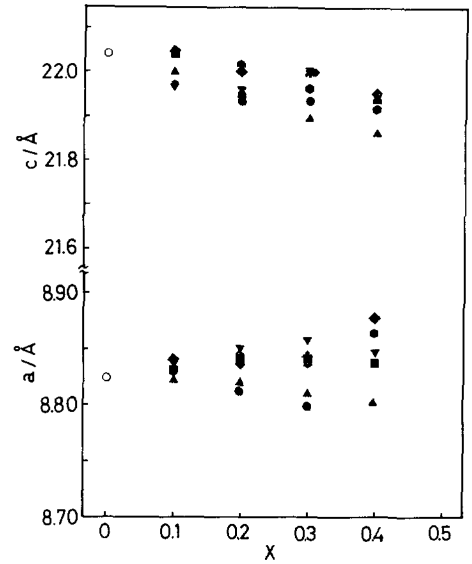

Fig. 8. Variation of the lattice constants on x for $Li_{1+x}M_xHf_{2-x}(PO_4)_3$, M=Cr (●); Fe(▲); Sc (▼); In(■); Lu (♦) or Y (●).

3.3. $Li_{1+x}M_xHf_{2-x}(PO_4)_3$ (M=Cr, Fe, Sc, In, Lu, or Y) system

In these systems, the samples were sintered at above 1293 K, and the sintering temperature was chosen in such a way that a maximum conductivity was obtained. The NASICON-type structure was formed for all the samples examined. A small amount of $P2_1/n$ monoclinic phase remained as a secondary phase for some samples. Fig. 8 presents the variation of lattice constants as a function of x for $Li_{1+x}M_xHf_{2-x}(PO_4)_3$ (M=Cr, Fe, Sc, In, Lu, or Y) systems. the rhombohedral phase distorted into monoclinic (space group C2/c) phase by the M^{3+} substitution for all the samples with $x=0.5$. The lattice constant of a-axis increased with the replacement of Hf^{4+} ion for larger M^{3+} ions (Sc^{3+}, In^{3+}, Lu^{3+} or Y^{3+}) and that decreased by smaller cations (Cr^{3+} or Fe^{3+}). However, c parameter decreased for all the $Li_{1+x}M_xHf_{2-x}(PO_4)_3$ systems examined. It is not clear why c parameter decreased by the M^{3+} ion substitution. The decrease of c-axis has also been reported for Zr^{4+}-systems of NASICON [2] and $Na_{1+x}M_xZr_{2-x}(PO_4)_3$ (M=Cr, In, Yb) [20]. On the other hand, for Ti^{4+}-system of $Li_{1+x}M_xTi_{2-x}(PO_4)_3$ (M=Al, Cr, Sc, etc.) [11], the lattice constant of c-axis was clearly increased by larger M^{3+} ion substitution. The HfO_6 and ZrO_6 octahedra in rhombohedral skeleton would be easy to distort because of a larger ionic radius for Hf^{4+} and Zr^{4+} ions. The oxygen coordination of HfO_6 may be shifted by the M^{3+} substitution or Li^+ insertion into A_2 sites.

The relationship between the porosity of the sintered pellet and x for $Li_{1+x}M_xHf_{2-x}(PO_4)_3$ system is shown in fig. 9. The high density pellets could be obtained by the x increase except for M=Cr system. The sintering temperature above 1393 K is necessary to obtain the high density samples for all the systems examined. Fig. 10 presents the relationship between the total conductivity at 298 K and x value for $Li_{1+x}M_xHf_{2-x}(PO_4)_3$ system. The conductivity increased greatly with x except for M=Cr system. A maximum conductivity was obtained for $x=0.2$ or 0.3, and the value was 1.7×10^{-4} S·cm^{-4} for the sample of $Li_{1.2}Fe_{0.2}Hf_{1.8}(PO_4)_3$. For M=Cr system, the conductivity and the sinterability did not increase with x. A very small amount of $Li_3Cr_2(PO_4)_3$ was formed as a secondary phase, which was confirmed by the X-ray measurement. However, $Li_3Cr_2(PO_4)_3$ is a very stable phosphate and has a

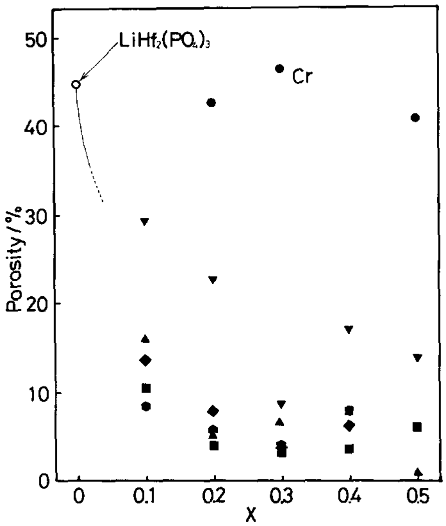

Fig. 9. Relationship between the porosity for the sintered pellet and x value for $Li_{1+x}M_xHf_{2-x}(PO_4)_3$, M=Cr (●); Fe (▲); Sc (▼); In (■); Lu (♦) or Y (●).

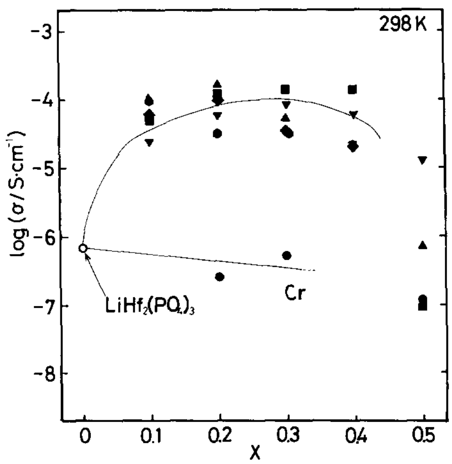

Fig. 10. The total conductivity at 298 K for $Li_{1+x}M_xHf_{2-x}(PO_4)_3$ M=Cr (●); Fe (▲); Sc (▼); In (■); Lu (♦) or Y (●).

high melting point [21]. The second phase might not act as a flux for a high density sample. Table 2 shows the activation energy of the bulk component for $Li_{1+x}M_xHf_{2-x}(PO_4)_3$ system. The activation energy could not be determined for the other samples, since the two minimum points on the Cole–Cole plot could not be clearly obtained. The activation energy for bulk component is also ca. 0.42 eV, which agreed with

Table 2
The activation energy of the bulk component for the $Li_{1+x}M_xHf_{2-x}(PO_4)_3$ system.

Sample	Bulk (eV)
$Li_{1.3}Fe_{0.3}Hf_{1.7}(PO_4)_3$	0.42
$Li_{1.3}Sc_{0.3}Hf_{1.7}(PO_4)_3$	0.42
$Li_{1.2}In_{0.2}Hf_{1.8}(PO_4)_3$	0.41
$Li_{1.3}In_{0.3}Hf_{1.7}(PO_4)_3$	0.43

that of $LiHf_2(PO_4)_3+yLi_2O$ system. The partial M^{3+} ion substitution and the Li^+ occupation at A_2 site did not influence the activation energy for Li^+ migration in the NASICON-type structure. The conductivity enhancement is not influenced by the M^{3+} ion substitution.

In conclusion, the electrical properties were studied for the ceramic electrolytes based on $LiHf_2(PO_4)_3$. The phase transition from $P2_1/n$ phase to NASICON-type $R\bar{3}c$ structure was observed at above 1173 K. The conductivity was greatly enhanced by the addition of the lithium compound and partial substitution of Hf^{4+} site by M^{3+} ion because of the high densification and the decrease in the activation energy at the grain boundary. The M^{3+} substitution did not affect the activation energy for bulk component. The activation energy for bulk component is 0.42 eV for $LiHf_2(PO_4)_3$ system, and is higher than 0.30 eV of $LiTi_2(PO_4)_3$ system. The tunnel size for a Li^+ migration of $LiHf_2(PO_4)_3$ system is larger than a suitable size. The maximum conductivity of 1.7×10^{-4} S·cm^{-1} at 298 K was obtained for presented $LiHf_2(PO_4)_3$ systems, and the value is smaller than 7.0×10^{-4} S·cm^{-1} [10,11] of $LiTi_2(PO_4)_3$ systems. However, the Hf-system is superior for the application such as battery and gas sensors, since the tetravalent Hf^{4+} cation is more stable toward a lithium metal and reductive gases than Ti^{4+} ion.

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